

UNIT: I

MATRICES & SYSTEM OF LINEAR EQUATIONS

Qm:

① Define Rank of matrix

Let A be a m by n matrix the rank of A is said to be r if

→ There is atleast 1 minor of A of order r which does not vanish

→ Every minor of A of order $(r+1)$ and higher order vanishes

In other words,

The rank of a matrix is the largest of the order of all the non-vanishing minors of that matrix.

② Define the system of linear equations (solution)

Set of values of $x_1, x_2, x_3, \dots, x_n$ which satisfy all the given m -equations in the system simultaneously is called the solution of system of linear equations.

③ Define Consistent & Inconsistent of system of equations

If the system of linear equation $Ax = B$ has the solution then it is consistent otherwise the system is inconsistent.

④ State Rouché's Theorem (State the necessary & sufficient condition of a system of linear non-homo eqn is to be consistent.

The system of linear equation $Ax = B$ is consistent if and only if the coefficient matrix A , the augmented matrix $[A, B]$ (or) $[A|B]$ or of the same rank.

$$(ie) . R[A] = R[A, B]$$

(5) What is the condition for the system with n linear equations to have

- Exactly one solution
- Infinitely many solution
- No solution

The given system of equation can be written as $Ax = B$

Then

- If $R[A] = R[A, B] = n$, then the system of equations are consistent & have unique solution.
- If $R[A] = R[A, B] < n$, then the system of equations are consistent & have infinitely many solutions
- If $R[A] \neq R[A, B]$, then the system of equations is inconsistent and has no solution.

8m:

6. Test the consistency of the following system of equations and solve if consistent

(i) $3x + y + z = 8$

$$-x + y - 2z = -5$$

$$x + y + z = 6$$

$$-2x + 2y - 3z = -7$$

Sol:

The above eqn can be written as

$$Ax = B$$

$$\text{Q6) } \begin{bmatrix} 3 & 1 & 1 \\ -1 & 1 & 1 \\ -2 & 2 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ -5 \\ 6 \end{bmatrix}$$

The augmented matrix is

$$[A, B] = \left[\begin{array}{ccc|c} 3 & 1 & 1 & 8 \\ -1 & 1 & -2 & -5 \\ 1 & 1 & 1 & 6 \\ -2 & 2 & -3 & -7 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ \textcircled{-1} & 1 & -2 & -5 \\ \textcircled{3} & 1 & 1 & 8 \\ \textcircled{-2} & \textcircled{2} & \textcircled{-3} & -7 \end{array} \right]$$

$$R_1 \leftrightarrow R_3$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 2 & -1 & 1 \\ 0 & -2 & -2 & -10 \\ 0 & 4 & -1 & 5 \end{array} \right]$$

$$R_2 \rightarrow R_2 + R_1$$

$$R_3 \rightarrow R_3 - 3R_1$$

$$R_4 \rightarrow R_4 + 2R_1$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 2 & -1 & 1 \\ 0 & 0 & -3 & -9 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

$$R_3 \rightarrow R_3 + R_2$$

$$R_4 \rightarrow R_4 - 2R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 2 & -1 & 1 \\ 0 & 0 & -3 & -9 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$R_4 \rightarrow 3R_4 + R_3$$

$$\rightarrow (A) \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 2 & -1 & 1 \\ 0 & 0 & -3 & -9 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

∴ It is an echelon form

It has 3 non-zero rows

$$\therefore R[A] = R[A, B] = A = 3$$

⇒ The system of equation is consistent & It has solution

$$R_1 \Rightarrow x + y + z = 6 \rightarrow (1)$$

$$R_2 \Rightarrow 2y - z = 1 \rightarrow (2)$$

$$-3z = -9 \rightarrow (3)$$

$$(3) \Rightarrow (z = 3)$$

sub in (2)

$$(2) \Rightarrow 2y - 3 = 1$$

$$2y = 1 + 3$$

$$2y = 4$$

$$(y = 2)$$

sub y & z in (1)

$$(1) \Rightarrow x + 2 + 3 = 6$$

$$x + 5 = 6$$

$$x = 6 - 5$$

$$(x = 1)$$

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$$(ii) 2x_1 + x_2 + 2x_3 + x_4 = 6$$

$$x_1 - x_2 + x_3 + 2x_4 = 6$$

$$4x_1 + 3x_2 + 3x_3 - 3x_4 = -1$$

$$2x_1 + 2x_2 + x_3 + x_4 = 10$$

Sol:

The above eqn can be written as

$$AX = B$$

$$\begin{bmatrix} 2 & 1 & 2 & 1 \\ 1 & -1 & 1 & 2 \\ 4 & 3 & 3 & -3 \\ 2 & 2 & -1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

$$\begin{bmatrix} 6 \\ 6 \\ -1 \\ 10 \end{bmatrix}$$

The augmented matrix is

$$[A, B] = \left[\begin{array}{cccc|c} 2 & 1 & 2 & 1 & 6 \\ 1 & -1 & 1 & 2 & 6 \\ 4 & 3 & 3 & -3 & -1 \\ 2 & 2 & -1 & 1 & 10 \end{array} \right]$$

$$\sim \left[\begin{array}{cccc|c} 1 & -1 & 1 & 2 & 6 \\ 2 & 1 & 2 & 1 & 6 \\ 4 & 3 & 3 & -3 & -1 \\ 2 & 2 & -1 & 1 & 10 \end{array} \right]$$

$$R_1 \leftrightarrow R_2$$

$$\sim \left[\begin{array}{cccc|c} 1 & -1 & 1 & 2 & 6 \\ 0 & 3 & 0 & -3 & -6 \\ 0 & 7 & -1 & -11 & -25 \\ 0 & 4 & -3 & -3 & -2 \end{array} \right]$$

$$R_2 \rightarrow R_2 - 2R_1$$

$$R_3 \rightarrow R_3 - 4R_1$$

$$R_4 \rightarrow R_4 - 2R_1$$

$$\sim \left[\begin{array}{cccc|c} 1 & -1 & 1 & 2 & 6 \\ 0 & 1 & 0 & -1 & -2 \\ 0 & 7 & -1 & -11 & -25 \\ 0 & 4 & -3 & -3 & -2 \end{array} \right]$$

$$R_2 \rightarrow \frac{R_2}{3}$$

$$\sim \left[\begin{array}{cccc|c} 1 & -1 & 1 & 2 & 6 \\ 0 & 1 & 0 & -1 & -2 \\ 0 & 0 & -1 & -4 & -11 \\ 0 & 0 & -3 & -3 & 6 \end{array} \right]$$

$$R_3 \rightarrow R_3 - 7R_2$$

$$R_4 \rightarrow R_4 - 4R_2$$

$$\sim \left[\begin{array}{cccc|c} 1 & -1 & 1 & 2 & 6 \\ 0 & 1 & 0 & -1 & -2 \\ 0 & 0 & -1 & -4 & -11 \\ 0 & 0 & 0 & -12 & 39 \end{array} \right]$$

$$R_4 \rightarrow R_4 - 3R_3$$

$\therefore$ It is an echelon form. It has 4 non-zero rows

$$\therefore R[A] = R[A, B] = n = 4$$

$\therefore$ The system is consistent & it has unique solutions.

$$\textcircled{A} \Rightarrow R_1 \Rightarrow x_1 - x_2 + x_3 + 2x_4 = 6 \rightarrow \textcircled{1}$$

$$R_2 \Rightarrow x_2 - x_4 = -2 \rightarrow \textcircled{2}$$

$$R_3 \Rightarrow -x_3 - 4x_4 = -11 \rightarrow \textcircled{3}$$

$$R_4 \Rightarrow 13x_4 = 39 \rightarrow \textcircled{4}$$

$$x_4 = 39/13$$

$$\boxed{x_4 = 3}$$

Sub in (3) & (2)

$$\textcircled{3} \Rightarrow -x_3 - 12 = -11$$

$$-x_3 = -11 + 12$$

$$-x_3 = 1$$

$$\boxed{x_3 = -1}$$

$$\textcircled{2} \Rightarrow x_2 - 3 = -2$$

$$x_2 = -2 + 3$$

$$\boxed{x_2 = 1}$$

$$\textcircled{1} \Rightarrow x_1 - 1 + 1 + 6 = 6$$

$$x_1 - 2 + 6 = 6$$

$$x_1 + 4 = 6$$

$$x_1 = 6 - 4$$

$$\boxed{x_1 = 2}$$

Sol :

$$x_1 = 2$$

$$x_2 = 1$$

$$x_3 = -1$$

$$x_4 = 3$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

For the system of linear equations to have a solution, the rank of the coefficient matrix must be equal to the rank of the augmented matrix.

$$\rho(A) = \rho([A|b]) = 4$$

The system of linear equations has a unique solution.

(iii)

$$2x + y + 5z + w = 5$$

$$x + y + 3z - 4w = -1$$

$$3x + 6y - 2z + w = 8$$

$$2x + 2y + 2z - 3w = 2$$

The above equation can be written as

$$AX = B$$

$$\begin{bmatrix} 2 & 1 & 5 & 1 \\ 1 & 1 & 3 & -4 \\ 3 & 6 & -2 & 1 \\ 2 & 2 & 2 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 5 \\ -1 \\ 8 \\ 2 \end{bmatrix}$$

The Augmented matrix

$$[A|B] = \left[\begin{array}{cccc|c} 2 & 1 & 5 & 1 & 5 \\ 1 & 1 & 3 & -4 & -1 \\ 3 & 6 & -2 & 1 & 8 \\ 2 & 2 & 2 & -3 & 2 \end{array} \right]$$

$$\sim \left[\begin{array}{cccc|c} 1 & 1 & 3 & -4 & -1 \\ 2 & 1 & 5 & 1 & 5 \\ 3 & 6 & -2 & 1 & 8 \\ 2 & 2 & 2 & -3 & 2 \end{array} \right] \begin{array}{l} R_1 \leftrightarrow R_2 \\ R_2 - R_1 \\ R_3 - 3R_1 \\ R_4 - 2R_1 \end{array}$$

$$\sim \left[\begin{array}{cccc|c} 1 & 1 & 3 & -4 & -1 \\ 0 & -1 & -1 & 7 & 3 \\ 0 & 3 & -11 & 11 & 5 \\ 0 & 0 & -4 & 5 & 0 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 3R_1 \\ R_4 \rightarrow R_4 - 2R_1 \end{array}$$

$$\sim \left[\begin{array}{cccc|c} 1 & 1 & 3 & -4 & -1 \\ 0 & -1 & -1 & 7 & 3 \\ 0 & 0 & -14 & 10 & -4 \\ 0 & 0 & 0 & 15 & 8 \end{array} \right] \begin{array}{l} R_3 \rightarrow R_3 + 3R_2 \\ R_4 \rightarrow R_4 - 4R_2 \\ R_4 \rightarrow R_4 - 7R_3 \end{array}$$

It is an echelon form It has 4 non-zero rows

$$\therefore R[A] = R[A, B] = n = 4$$

$\therefore$ The system is consistent & It has unique solution.

$$\textcircled{1} \Rightarrow R_1 \Rightarrow x + y + 3z - 4w = -1 \rightarrow \textcircled{1}$$

$$R_2 \Rightarrow -y - z + 7w = 3 \rightarrow \textcircled{2}$$

$$R_3 \Rightarrow -14z + 10w = -4 \rightarrow \textcircled{3}$$

$$R_4 \Rightarrow 15w = 8 \rightarrow \textcircled{4}$$

$$\boxed{w = 8/15}$$

Sub in (3)

$$\textcircled{3} \Rightarrow -y - 14z + 10 \times \frac{8}{15} = -4$$

$$-14z + \frac{80}{15} = -4$$

$$-14z = -4 - \frac{80}{15}$$

$$-14z = \frac{-140}{15}$$

$$z = \frac{-140}{15 \times -14}$$

$$\boxed{z = 2/3}$$

Sub in (2)

$$\textcircled{2} \Rightarrow -y - \frac{2}{3} + 7 \times \frac{8}{15} = 3$$

$$-y - \frac{2}{3} + \frac{56}{15} = 3$$

$$-y - \frac{10 + 56}{15} = 3$$

$$-y + \frac{46}{15} = 3$$

$$-y = 3 - \frac{46}{15} = \frac{7}{15}$$

$$\boxed{y = 7/15}$$

Sub in (1)

$$\textcircled{1} \Rightarrow x + \frac{1}{15} + 3 \times \frac{2}{3} - 4 \times \frac{8}{15} = -1$$

$$x + \frac{1}{15} + \frac{6}{3} - \frac{32}{15} = -1$$

$$x + \frac{1+30-32}{15} = -1$$

$x \times$

$$x - \frac{1}{15} = -1$$

$$x = -1 + \frac{1}{15}$$

$$\boxed{x = \frac{-14}{15}}$$

Sol:

$$x = -14/15$$

$$y = 1/15$$

$$z = 2/3$$

$$w = 8/15$$

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7) Test the consistency of the following system of equations & solve it if it is consistent.

$$\textcircled{1} \quad x - 4y - 3z + 16 = 0$$

$$2x + 7y + 12z = 48$$

$$4x - y + 6z = 16$$

$$5x - 5y + 3z = 0$$

Sol: The given equation is written as

$$AX = B$$

$$\begin{bmatrix} 1 & -4 & -3 \\ 2 & 7 & 12 \\ 4 & -1 & 6 \\ 5 & -5 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -16 \\ 48 \\ 16 \\ 0 \end{bmatrix}$$

The Augmented matrix is

$$[A, B] = \left[\begin{array}{ccc|c} 1 & -4 & -3 & -16 \\ 2 & 7 & 12 & 48 \\ 4 & -1 & 6 & 16 \\ 5 & -5 & 3 & 0 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & -4 & -3 & -16 \\ 0 & 15 & 18 & 80 \\ 0 & 15 & 18 & 80 \\ 0 & 15 & 18 & 80 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 4R_1 \\ R_4 \rightarrow R_4 - 3R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & -4 & -3 & -16 \\ 0 & 15 & 18 & 80 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \begin{array}{l} R_3 \rightarrow R_3 - R_2 \\ R_4 \rightarrow R_4 - R_2 \end{array}$$

$\rightarrow \textcircled{A}$

$\therefore$ It is an echelon form, It has ² non-zero rows

$$\therefore R[A] = R[A, B] = 2 < n = 3$$

$\therefore$ The system of equation is consistent & it has infinite number of solutions.

$$\textcircled{A} \Rightarrow R_1 \Rightarrow x - 4y - 3z = -16 \rightarrow \textcircled{1}$$

$$R_2 \Rightarrow 15y + 18z = 80 \rightarrow \textcircled{2}$$

Let $z = k \in \mathbb{R}$, in $\textcircled{2}$

$$\textcircled{2} \Rightarrow 15y + 18k = 80$$

$$15y = \frac{80 - 18k}{15}$$

$$y = \frac{80}{15} - \frac{18k}{15}$$

$$y = \frac{16}{3} - \frac{6}{5}k //$$

$$\left[\begin{array}{ccc|c} 1 & -4 & -3 & -16 \\ 0 & 15 & 18 & 80 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\textcircled{1} \Rightarrow x - 4 \left[\frac{16}{3} - \frac{6k}{5} \right] - 3k = -16$$

$$x - \frac{64}{3} + \frac{24k}{5} - 3k = -16$$

$$x + \frac{24k - 15k}{15} = -16 + \frac{64}{3}$$

$$x + \frac{9k}{5} = \frac{-48 + 64}{3}$$

$$x = \frac{16}{3} - \frac{9k}{5} //$$

Sol:

$$x = \frac{16}{3} - \frac{9k}{5}$$

$$y = \frac{16}{3} - \frac{6k}{5}$$

$$z = k$$

ii)

$$x_1 + 2x_2 - x_3 - 5x_4 = 4$$

$$x_1 + 3x_2 - 2x_3 - 7x_4 = 5$$

$$2x_1 - x_2 + 3x_3 = 3$$

Sol: The given equation is written as $X \cdot AX = B$

$$\begin{bmatrix} 1 & 2 & -1 & -5 \\ 1 & 3 & -2 & -7 \\ 2 & -1 & 3 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 3 \end{bmatrix}$$

The Augmented matrix is

$$[A, B] = \left[\begin{array}{cccc|c} 1 & 2 & -1 & -5 & 4 \\ 1 & 3 & -2 & -7 & 5 \\ 2 & -1 & 3 & 0 & 3 \end{array} \right]$$

$$\sim \left[\begin{array}{cccc|c} 1 & 2 & -1 & -5 & 4 \\ 0 & 1 & -1 & -2 & 1 \\ 0 & -5 & 5 & 10 & -5 \end{array} \right]$$

$$R_2 \rightarrow R_2 - R_1 \quad R_3 \rightarrow R_3 - 2R_1$$

$$R_3 \rightarrow R_3 - 2R_1$$

$$\sim \left[\begin{array}{cccc|c} 1 & 2 & -1 & -5 & 4 \\ 0 & 1 & -1 & -2 & 1 \\ 0 & -1 & 1 & 2 & -1 \end{array} \right]$$

$$R_3 \rightarrow R_3 / 5$$

$$\sim \left[\begin{array}{cccc|c} 1 & 2 & -1 & -5 & 4 \\ 0 & 1 & -1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 + R_2} \rightarrow \textcircled{A}$$

$\therefore$ It is an echelon form. It has 2 non-zero

$$\therefore R[A] = R[A, B] = 2 < n = 4$$

$\therefore$ The system of equation, are consistent & it has an infinity of solution.

$$\textcircled{A} \Rightarrow R_1 \Rightarrow x_1 + 2x_2 - x_3 - 5x_4 = 4 \rightarrow \textcircled{1}$$

$$R_2 \Rightarrow x_2 - x_3 - 2x_4 = 1 \rightarrow \textcircled{2}$$

$$\text{Let } x_3 = k_1, x_4 = k_2, \in \mathbb{R}$$

$$\textcircled{2} \Rightarrow x_2 - k_1 - 2k_2 = 1$$

$$x_2 = 1 + k_1 + 2k_2$$

$$x_1 + 2 + 2k_1 + 4k_2 - k_1 - 5k_2 = 4 \Rightarrow x_1 + k_1 - k_2 = 2$$

$$x_1 + k_1 - k_2 = 2 \Rightarrow x_1 = 2 - k_1 + k_2$$

Sol:

$$\begin{cases} x_1 = 2 - k_1 + k_2 \\ x_2 = 1 + k_1 + 2k_2 \\ x_3 = k_1 \\ x_4 = k_2 \end{cases}$$

$$x_1 = 2 - k_1 + k_2$$

$$x_2 = 1 + k_1 + 2k_2$$

$$x_3 = k_1$$

$$x_4 = k_2$$

$$\forall k_1, k_2 \in \mathbb{R}$$

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8) Test the consistency of following system of equations

$$x - 2y - 3z = 2$$

$$3x - 2y - z = 1$$

$$-2y - 3z = 2$$

$$y + 2z = 1$$

$$\left[\begin{array}{ccc|c} 1 & -2 & -3 & 2 \\ 3 & -2 & -1 & 1 \\ 0 & -2 & -3 & 2 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & -2 & -3 & 2 \\ 0 & 4 & 8 & -5 \\ 0 & -2 & -3 & 2 \end{array} \right]$$

Sol: The Augmented matrix

$$[A, B] = \left[\begin{array}{ccc|c} 1 & -2 & -3 & 2 \\ 3 & -2 & 0 & -1 \\ 0 & -2 & -3 & 2 \\ 0 & 1 & 2 & 1 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & -2 & -3 & 2 \\ 0 & 4 & 9 & -7 \\ 0 & -2 & -3 & -2 \\ 0 & 1 & 2 & 1 \end{array} \right] \quad R_2 \rightarrow R_2 - 3R_1$$

$$\sim \left[\begin{array}{ccc|c} 1 & -2 & -3 & 2 \\ 0 & 4 & 9 & -7 \\ 0 & 0 & 3 & 9 \\ 0 & 0 & -1 & -4 \end{array} \right] \quad \begin{array}{l} R_3 \rightarrow 2R_3 + R_2 \\ R_4 \rightarrow 4R_4 - R_2 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & -2 & -3 & 2 \\ 0 & 4 & 9 & -7 \\ 0 & 0 & 3 & -3 \\ 0 & 0 & 0 & 30 \end{array} \right] \quad R_4 \rightarrow 3R_4 + R_3$$

$\therefore$ It is an echelon form

$\therefore$ Here $R[A] = 3 \neq R[A, B] = 4$

$\therefore$ The system of equation is inconsistent and has no solution

(ii) $x_1 + x_2 + x_3 = 1$

$2x_1 - 2x_2 + 3x_3 = 1$

$x_1 - x_2 + 2x_3 = 5$

$x_1 + x_2 + x_3 = 2$

The given equation can be written as $AX = B$

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & -2 & 3 \\ 1 & -1 & 2 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 5 \\ 2 \end{bmatrix}$$

The Augmented Matrix is

$$[A, B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 2 & -2 & 3 & 1 \\ 1 & -1 & 2 & 5 \\ 1 & 1 & 1 & 2 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -4 & 1 & -1 \\ 0 & -2 & 1 & 4 \\ 0 & 0 & 0 & 1 \end{array} \right]$$

$$R_2 \rightarrow R_2 - 2R_1$$

$$R_3 \rightarrow R_3 - R_1$$

$$R_4 \rightarrow R_4 - R_1$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -4 & 1 & -1 \\ 0 & 0 & 1 & 9 \\ 0 & 0 & 0 & 1 \end{array} \right]$$

$$R_3 \rightarrow 2R_3 - R_2$$

$$R[A] = 3 \quad R[A, B] = 4$$

$$R[A] \neq R[A, B]$$

The system of equation is inconsistent
and it has no solution.

9) (i) Find the values of k for which the equations

$$x + y + z = 1$$

$$x + 2y + 4z = k$$

$$x + 4y + 10z = k^2 \quad \text{may have solutions.}$$

For these values of k find the solution also.

The given eqn can be written as

$$AX = B$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 4 & 10 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ k \\ k^2 \end{bmatrix}$$

The Augmented matrix

$$[A, B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & k \\ 1 & 4 & 10 & k^2 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & k-1 \\ 0 & 3 & 9 & k^2-1 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & k-1 \\ 0 & 0 & 0 & k^2+3k+2 \end{array} \right] R_3 \rightarrow R_3 - 3R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & k-1 \\ 0 & 0 & 0 & (k-1)(k-2) \end{array} \right] \rightarrow \textcircled{A}$$

It is an echelon form

Here $R(A) = 2$, $\Rightarrow R[A, B]$ also 2

So $(k-1)(k-2) = 0$ if $k=1$ or $k=2$

$$\boxed{k=1, 2}$$

(i) Put $k=1$ in $\textcircled{A} \Rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$

$$x+y+z=1 \rightarrow \textcircled{1}$$

$$y+3z=0 \rightarrow \textcircled{2}$$

Put $z=t \in \mathbb{R}$ in $\textcircled{2}$

$$\textcircled{2} \Rightarrow y+3t=0$$

$$\boxed{y = -3t} \text{ Sub in } \textcircled{1}$$

$$\textcircled{1} \Rightarrow x - 3t + t = 1$$

$$\boxed{x = 1 + 2t}$$

$$(x, y, z) = [1 + 2t; -3t; t]; \forall t \in \mathbb{R}$$

(ii) Put $k=2$ in (A)

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$x + y + z = 1 \Rightarrow (3)$$

$$y + 3z = 1 \Rightarrow (4)$$

Put $z = s \in \mathbb{R}$, in (4)

$$y + 3s = 1$$

$$\boxed{y = 1 - 3s}$$

Sub in (3)

$$x + 1 - 3s + s = 1$$

$$x - 2s = 0$$

$$\boxed{x = 2s}$$

$$(x, y, z) = [2s; 1 - 3s; s]; \forall s \in \mathbb{R}$$

(ii) Find the values of the λ for which the equations

$$x + 2y + z = 4$$

$$2x - y - z = +3\lambda$$

$$4x - 7y - 5z = \lambda^2 \text{ may have solutions.}$$

For these values of λ find the solution also

Sol:

The given eqn can be written as $Ax = B$

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & -1 \\ 4 & -7 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 3\lambda \\ \lambda^2 \end{bmatrix}$$

$$\boxed{Ax = B}$$

$$1 = 3 + 3s - x \Rightarrow (1)$$

$$\boxed{3s + 1 = x}$$

The Augmented matrix

$$[A, B] = \left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 2 & -1 & -1 & 3\lambda \\ 4 & -7 & -5 & \lambda^2 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & -5 & -3 & 3\lambda - 8 \\ 0 & -15 & -9 & \lambda^2 - 16 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 4R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & -5 & -3 & 3\lambda - 8 \\ 0 & 0 & 0 & \lambda^2 - 9\lambda + 8 \end{array} \right] R_3 \rightarrow R_3 - 3R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & -5 & -3 & 3\lambda - 8 \\ 0 & 0 & 0 & (\lambda - 1)(\lambda - 8) \end{array} \right] \rightarrow \textcircled{A}$$

$\therefore$ It is an echelon form

Here $R(A) = 2 \Rightarrow R[A, B] = 2$

and given that the system may have solution.

Hence by Rouches theorem

$$R[A, B] = 2$$

which is possible only if $(\lambda - 1)(\lambda - 8) = 0 \Rightarrow \boxed{\lambda = 1, 8}$

$\therefore$ The system has solution in two cases.

(i) put $\lambda = 1$ in (A)

$$\textcircled{A} \Rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & -5 & -3 & -5 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & -5 & -3 & -5 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$R_1 \Rightarrow x + 2y + z = 4 \rightarrow (1)$$

$$R_2 \Rightarrow -5y - 3z = -5 \rightarrow (2)$$

Put $z = t \in \mathbb{R}$, in eqn (2)

$$(2) \Rightarrow -5y - 3t = -5$$

$$-5y = -5 + 3t$$

$$y = \frac{-5 + 3t}{-5}$$

$$\boxed{y = \frac{5 - 3t}{5}}$$

Sub y in (1)

$$(1) \Rightarrow x + 2 \left[\frac{5 - 3t}{5} \right] + t = 4$$

$$x + \frac{10}{5} - \frac{6t}{5} + t = 4$$

$$x + 2 - \frac{6t}{5} + t = 4$$

$$x = \frac{6t}{5} + t = 4 - 2$$

$$x = \frac{6t + 5t}{5} = 2$$

$$x = \frac{t}{5} = 2$$

$$x = 2 + \frac{t}{5}$$

$$\boxed{x = \frac{t + 10}{5}}$$

$$(x, y, z) = \left[\frac{t + 10}{5} ; \frac{5 - 3t}{5} ; t \right], \forall t \in \mathbb{R}$$

(ii) Put $\lambda = 8$ in (A)

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & -5 & -3 & 16 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$R_1 \Rightarrow x + 2y + z = 4 \rightarrow (3)$$

$$R_2 \Rightarrow -5y - 3z = 16 \rightarrow (4)$$

put $z = s \in \mathbb{R}$, in equ (4)

$$(4) \Rightarrow -5y - 3s = 16$$

$$-5y = 16 + 3s$$

$$y = \frac{16 + 3s}{-5} \quad (\text{or})$$

$$\boxed{\frac{-16 - 3s}{5}}$$

sub y in (3)

$$(3) \Rightarrow x + 2 \left[\frac{-16 - 3s}{5} \right] + s = 4$$

$$x - \frac{32}{5} - \frac{6s}{5} + s = 4$$

$$x - \frac{6s}{5} + s = 4 + \frac{32}{5}$$

$$x - \frac{6s + 5s}{5} = 4 + \frac{32}{5}$$

$$x - \frac{s}{5} = 4 + \frac{32}{5}$$

$$x - \frac{s}{5} = \frac{20 + 32}{5}$$

$$x - \frac{s}{5} = \frac{52}{5} \Rightarrow x = \frac{52}{5} + \frac{s}{5}$$

$$\boxed{x = \frac{52 + s}{5}} \quad (\text{or})$$

$$(x, y, z) = \left[\frac{52 + s}{5} ; \frac{-16 - 3s}{5} ; s \right]; \forall s \in \mathbb{R}.$$

10) (i) Find the values of k for which the equations

$$\begin{aligned} kx + y + z &= 1 \\ x + ky + z &= 1 \\ x + y + kz &= 1 \end{aligned} \text{ have}$$

- infinitely many solutions
- no solutions
- unique solutions.

Sol: The given eqn can be written as $AX = B$

$$\begin{bmatrix} k & 1 & 1 \\ 1 & k & 1 \\ 1 & 1 & k \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

The Augmented matrix is

$$[A, B] = \left[\begin{array}{ccc|c} k & 1 & 1 & 1 \\ 1 & k & 1 & 1 \\ 1 & 1 & k & 1 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|c} k & 1 & 1 & 1 \\ 0 & 1-k^2 & 1-k & 1-k \\ \dots & \dots & \dots & \dots \end{array} \right]$$

$R_1 \leftrightarrow R_2$

$$\sim \left[\begin{array}{ccc|c} 1 & k & 1 & 1 \\ k & 1 & 1 & 1 \\ 1 & 1 & k & 1 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & k & 1 & 1 \\ 0 & 1-k^2 & 1-k & 1-k \\ 0 & 1-k & k-1 & 0 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & k-1 & 1 & 1 \\ 0 & 1-k^2 & 1-k & 1-k \\ 0 & 1-k & k-1 & 0 \end{array} \right]$$

$$\sim \begin{bmatrix} 1 & k & 1 \\ 0 & 1-k^2 & 1-k \\ 0 & 0 & 0 \end{bmatrix}$$

(or) $\frac{(k+2)(k-1)}{k+1}$

$$R_3 - \frac{1}{1+k} R_2 \\ (1-k) - \frac{1}{1+k} (1-k^2) \\ = (1-k) - (1-k) \\ = 0.$$

$$\sim \begin{bmatrix} 1 & k & 1 & 1 \\ 0 & 1-k^2 & 1-k & 1-k \\ 0 & 0 & k^2+k-2 & k-1 \end{bmatrix}$$

$$R_3 \rightarrow (1+k)R_3 - R_2$$

$$(1+k)(k-1) - (1-k)$$

$$k - 1 + k^2 - k - 1 + k \\ k^2 + k - 2$$

$$R_3 - \frac{1}{1+k} R_2$$

$$(k-1) - \frac{1}{1+k} (1-k) \\ = (k-1) + \frac{k-1}{1+k}$$

$$\sim \begin{bmatrix} 1 & k & 1 & 1 \\ 0 & 1-k^2 & 1-k & 1-k \\ 0 & 0 & (k+2)(k-1) & k-1 \end{bmatrix}$$

$$= (k-1) \left(1 + \frac{1}{1+k} \right) \\ = (k-1) \frac{2k-1}{1+k}$$

$\therefore$ It is an echelon form.

$$\begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

(i) If $k=1$, Then $R(A) = R(A|B) = 1 < n = 3$

$\therefore$ The system of equations has infinite no. of solutions.

(ii) If $k=-2$ Then $R(A) = 2 \neq R(A|B) = 3$

$\therefore$ The system of equations has no solution

(iii) If $k \neq 1, k \neq -2$, then $R(A) = R(A|B) = n = 3$

$\therefore$ The system of equation has unique solution.

(iv) for what values of λ, μ this the system of equations

$$x + y + 2z = 2$$

$$2x - y + 3z = 2$$

$$5x - y + \lambda z = \mu \text{ have}$$

- Infinitely many solutions
- no solutions
- unique solutions.

Sol: The given eqn can be written as $AX = B$

$$\begin{bmatrix} 1 & 1 & 2 \\ 2 & -1 & 3 \\ 5 & -1 & \lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ \mu \end{bmatrix}$$

The Augmented matrix

$$[A, B] = \left[\begin{array}{ccc|c} 1 & 1 & 2 & 2 \\ 2 & -1 & 3 & 2 \\ 5 & -1 & \lambda & \mu \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 2 & 2 \\ 0 & -3 & -1 & -2 \\ 0 & -6 & \lambda-10 & \mu-10 \end{array} \right]$$

$$\begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 5R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 2 & 2 \\ 0 & -3 & -1 & -2 \\ 0 & 0 & \lambda-8 & \mu-6 \end{array} \right]$$

$$\begin{array}{l} R_3 \rightarrow R_3 + 2R_2 \\ \lambda-10+2 \\ \mu-10+4 \end{array}$$

$\therefore$ It is in echelon form.

(i) If $\lambda = 8$ and $\mu = 6$, then $R[A] = R[A, B] = 2 < n = 3$

$\therefore$ The system has infinite number of solutions.

(ii) If $\lambda = 8$ and $\mu \neq 6$, then $R[A] = 2 \neq R[A, B] = 3$

$\therefore$ The system has no solution.

(iii) If $\lambda \neq 8$ and $\mu = \text{any value}$ then $R[A] = R[A, B] = n = 3$

$\therefore$ The system has unique solution.

(iii) Find the values of k for which the equations

$$kx - 2y + z = 1$$

$$x - 2ky + z = -2$$

$$x - 2y + kz = 1$$

have

- (i) infinitely many solutions
- (ii) no solutions
- (iii) unique solutions

The given eqn can be written as $AX = B$

$$\begin{bmatrix} K & -2 & 1 \\ 1 & -2k & 1 \\ 1 & -2 & k \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

The Augmented matrix

$$[A, B] = \left[\begin{array}{ccc|c} K & -2 & 1 & 1 \\ 1 & -2k & 1 & -2 \\ 1 & -2 & k & 1 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & -2k & 1 & -2 \\ K & -2 & 1 & 1 \\ 1 & -2 & k & 1 \end{array} \right] R_1 \leftrightarrow R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & -2k & 1 & -2 \\ 0 & +2k^2-2 & 1-k & 1+2k \\ 0 & +2k-2 & k-1 & 3 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - kR_1 \\ R_3 \rightarrow R_3 - R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & -2k & 1 & -2 \\ 0 & +2(k^2-1) & 1-k & 1+2k \\ 0 & 2(k-1) & k-1 & 3 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & -2k & 1 & -2 \\ 0 & 2(k^2-1) & 1-k & 1+2k \\ 0 & 0 & -k^2-k+2 & -k-2 \end{array} \right]$$

$\therefore$ It is in echelon form

(i) If $K = -2$, then $R[A] = R[A, B] = 2 < n = 3$

$\therefore$ The system has infinite number of solution

(ii) If $K = 1$, then $R[A] = 1 \neq R[A, B] = 3$

$\therefore$ The system has no solution

(iii) If $K \neq 1, -2, -1$, then $R[A] = R[A, B] = n = 3$

$\therefore$ The system has unique solution

(iv) For what values of λ and μ thus the system of equation

$$x + y + z = 6$$

$$x + 2y + 3z = 10$$

$$x + 2y + \lambda z = \mu$$

(a) infinitely many solution

(b) no solution

(c) unique solution

The given eqn can be written as $AX=B$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & \lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \\ \mu \end{bmatrix}$$

The Augmented matrix

$$[A|B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & \mu \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & \lambda-1 & \mu-6 \end{array} \right]$$

$$R_2 \rightarrow R_2 - R_1$$

$$R_3 \rightarrow R_3 - R_1$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & \lambda-3 & \mu-10 \end{array} \right]$$

$$R_3 \rightarrow R_3 - R_2$$

It is an echelon form.

(i) If $\lambda = 3$, $\mu = 10$, then $R(A) = R(A|B) = 2 < n = 3$

$\therefore$ The system has infinite number of solution.

(ii) If $\lambda = 3$, $\mu \neq 10$, then $R(A) = 2 \neq R(A|B) = 3$

$\therefore$ The system has no solution.

(iii) If $\lambda \neq 3$, $\mu = \text{any value}$ then $R(A) = R(A|B) = n = 3$

$\therefore$ The system has unique solution.

$$x = 6 - y - z$$

$$0 = 4 + 2y + 3z$$

$$-4 = 2y + 3z$$

11) Find the non-trivial solutions of the following systems of equations if it exists.

$$\begin{aligned} \text{(i)} \quad & x - y + 2z - 3w = 0 \\ & 3x + 2y - 4z + w = 0 \\ & 5x - 3y + 2z + 6w = 0 \\ & x - 9y + 14z - 2w = 0 \end{aligned}$$

Sol: The given eqn can be written as $AX = B$

$$\begin{bmatrix} 1 & -1 & 2 & -3 \\ 3 & 2 & -4 & 1 \\ 5 & -3 & 2 & 6 \\ 1 & -9 & 14 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

The Augmented matrix

$$[A, 0] = \left[\begin{array}{cccc|c} 1 & -1 & 2 & -3 & 0 \\ 3 & 2 & -4 & 1 & 0 \\ 5 & -3 & 2 & 6 & 0 \\ 1 & -9 & 14 & -2 & 0 \end{array} \right]$$

$$\sim \left[\begin{array}{cccc|c} 1 & -1 & 2 & -3 & 0 \\ 0 & 5 & -10 & 10 & 0 \\ 0 & 2 & -8 & 21 & 0 \\ 0 & -8 & 12 & 1 & 0 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - 3R_1 \\ R_3 \rightarrow R_3 - 5R_1 \\ R_4 \rightarrow R_4 - R_1 \end{array}$$

$$\sim \left[\begin{array}{cccc|c} 1 & -1 & 2 & -3 & 0 \\ 0 & 1 & -2 & 2 & 0 \\ 0 & 2 & -8 & 21 & 0 \\ 0 & -8 & 12 & 1 & 0 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 \cdot 5 \\ R_3 \rightarrow R_3 - 2R_2 \\ R_4 \rightarrow R_4 + 8R_2 \end{array}$$

$$\sim \left[\begin{array}{cccc|c} 1 & -1 & 2 & -3 & 0 \\ 0 & 1 & -2 & 2 & 0 \\ 0 & 0 & -4 & 17 & 0 \\ 0 & 0 & -4 & 17 & 0 \end{array} \right] \begin{array}{l} R_3 \rightarrow R_3 - 2R_2 \\ R_4 \rightarrow R_4 + 8R_2 \end{array}$$

$$\sim \left[\begin{array}{cccc|c} 1 & -1 & 2 & -3 & 0 \\ 0 & 1 & -2 & 2 & 0 \\ 0 & 0 & -4 & 17 & 0 \\ 0 & 0 & 0 & 24 & 0 \end{array} \right] \begin{array}{l} R_4 \rightarrow R_4 \div 24 \end{array}$$

∴ It is an echelon form

$|A| = 0$, Expanding along R_4

∴ The system has non-trivial solution

(A) $\Rightarrow$

$$R_1 \Rightarrow x - y + 2z - 3w = 0 \Rightarrow (1)$$

$$R_2 \Rightarrow y - 2z + 2w = 0 \Rightarrow (2)$$

$$R_3 \Rightarrow -4z + 17w = 0 \Rightarrow (3)$$

put $w = 4t, t \in \mathbb{R}$

$$(3) \Rightarrow -4z + 17(4t) = 0$$

$$4z = 17(4t)$$

$$\boxed{z = 17t}$$

$$(2) \Rightarrow y - 2(17t) + 2(4t) = 0$$

$$y - 34t + 8t = 0$$

$$y - 26t = 0$$

$$\boxed{y = 26t}$$

$$(1) \Rightarrow x - 26t + 2(17t) - 3(4t) = 0$$

$$x - 26t + 34t - 12t = 0$$

$$x - 38t + 34t = 0$$

$$x - 4t = 0$$

$$\boxed{x = 4t}$$

$$\therefore (x, y, z, w) = (4t; 26t; 17t; 4t); \forall t \in \mathbb{R}$$

(ii)

$$x + 2y + z + 2w = 0$$

$$x + 3y + 2z + 2w = 0$$

$$2x + 4y + 3z + 6w = 0$$

$$3x + 7y + 4z + 6w = 0$$

$$\begin{aligned} x &= 4t \\ y &= 26t \\ z &= 17t \\ w &= 4t \end{aligned}$$

The given system can be written as $AX = B$

$$\begin{bmatrix} 1 & 2 & 1 & 2 \\ 1 & 3 & 2 & 2 \\ 2 & 4 & 3 & 6 \\ 3 & 7 & 4 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

The Augmented matrix

$$\left[\begin{array}{cccc|c} 1 & 2 & 1 & 2 & 0 \\ 1 & 3 & 2 & 2 & 0 \\ 2 & 4 & 3 & 6 & 0 \\ 3 & 7 & 4 & 6 & 0 \end{array} \right]$$

$$\xrightarrow{R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - 2R_1, R_4 \rightarrow R_4 - 3R_1} \left[\begin{array}{cccc|c} 1 & 2 & 1 & 2 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{array} \right]$$

$$\xrightarrow{R_4 \rightarrow R_4 - R_2} \left[\begin{array}{cccc|c} 1 & 2 & 1 & 2 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \rightarrow \textcircled{A}$$

$$R_4 \rightarrow R_4 - R_2 \Rightarrow (0, 1, 1, 0) - (0, 1, 1, 0) = (0, 0, 0, 0)$$

$\therefore$ It is an echelon form

$|A| = 0$, Expanding along R_4

$\therefore$ The system has non-trivial solution

$\textcircled{A} \Rightarrow x$

$R_1 \Rightarrow x + 2y + z + 2w = 0 \rightarrow \textcircled{1}$

$R_2 \Rightarrow y + z = 0 \rightarrow \textcircled{2}$

$R_3 \Rightarrow z + 2w = 0 \rightarrow \textcircled{3}$

put $w = t, t \in \mathbb{R}$.

$\textcircled{3} \Rightarrow z + 2t = 0$
 $\boxed{z = -2t}$

Sub in ②

$$\textcircled{2} \Rightarrow y + z = 0$$

$$y - 2t = 0$$

$$\boxed{y = 2t}$$

Sub y, z, w in ①

$$\textcircled{1} \Rightarrow x + 2(2t) - 2t + 2t = 0$$

$$x + 4t = 0$$

$$\boxed{x = -4t}$$

$$\therefore (x, y, z, w) = [-4t; 2t; -2t; t] ; \forall t \in \mathbb{R}$$

$$\textcircled{1} \leftarrow 0 = w + 5 + 2t + 2t \leftarrow \textcircled{A}$$

$$\textcircled{2} \leftarrow 0 = 5 + 2t \leftarrow \textcircled{B}$$

$$\textcircled{3} \leftarrow 0 = w + 5 - 3t \leftarrow \textcircled{C}$$

$$\textcircled{4} \leftarrow 0 = 5 - 3t \leftarrow \textcircled{D}$$

$$\boxed{3t = 5}$$

2m:

12) Find A and B if the equations

$$2x - 3y = 5$$

$$ax + by = -10 \text{ have many solutions}$$

The given eqn can be written as $AX = B$

$$\begin{bmatrix} 2 & -3 \\ a & b \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ -10 \end{bmatrix}$$

The Augmented matrix.

$$[A, B] = \left[\begin{array}{cc|c} 2 & -3 & 5 \\ a & b & -10 \end{array} \right]$$

$$\sim \left[\begin{array}{cc|c} 2 & -3 & 5 \\ 0 & 2b+3a & -20-5a \end{array} \right] \quad R_2 \rightarrow 2R_2 - aR_1$$

It is a echelon form.

Given the system has many solution

$$\Rightarrow R(A) = R(A, B) < n = 2$$

$$\text{Hence } R(A) = R(A, B) = 1$$

It is possible only if

$$2b + 3a = 0 \rightarrow (1)$$

$$-20 - 5a = 0 \rightarrow (2)$$

$$(2) \rightarrow -5a = 20$$

$$\boxed{a = -4}$$

sub in (1)

$$(1) \rightarrow 2b - 12 = 0$$

$$2b = 12$$

$$\boxed{b = 6} //$$

13) Find the condition satisfied by a, b, c, so that the system of equations

$$x + 2y - 3z = a$$

$$3x - y + 2z = b$$

$$x - 5y + 8z = c$$

they have a solution

The given eqn can be written as $AX = B$

$$\begin{bmatrix} 1 & 2 & -3 \\ 3 & -1 & 2 \\ 1 & -5 & 8 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

The Augmented matrix

$$[A, B] = \left[\begin{array}{ccc|c} 1 & 2 & -3 & a \\ 3 & -1 & 2 & b \\ 1 & -5 & 8 & c \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & -3 & a \\ 0 & -7 & 11 & b-3a \\ 0 & -7 & 11 & c-a \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - 3R_1 \\ R_3 \rightarrow R_3 - R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & -3 & a \\ 0 & -7 & 11 & b-3a \\ 0 & 0 & 0 & 2a-b+c \end{array} \right] R_3 \rightarrow R_3 - R_2$$

$\therefore$ It is a echelon form

Given the system have solution

$$\Rightarrow R(A) = R(A, B)$$

$$\text{Here } R(A) = 2 \Rightarrow R(A, B) = 2$$

It is possible only if

$$2a - b + c = 0$$

$$\Rightarrow \boxed{b = 2a + c}$$

14. Write the algorithm for Gauss Jordan method.

This method is modification of Gauss elimination method. In this method the coefficient matrix A of the system of equation $Ax = B$ is to be diagonal matrix or unit matrix. By making the matrix A not only upper triangular matrix but also a lower triangular matrix by making all the matrix above the leading diagonal of matrix A are also zeroes.

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{array} \right] \begin{array}{l} x \\ y \\ z \end{array} = \begin{array}{l} a \\ b \\ c \end{array}$$

15) Write the diagonal dominance rule for a system of three linear equations with 3 unknowns.

sol: consider the system of equations

$$a_1x + b_1y + c_1z = d_1$$

$$a_2x + b_2y + c_2z = d_2$$

$$a_3x + b_3y + c_3z = d_3$$

let us assume,

$$|a_1| > |b_1| + |c_1|$$

$$|b_2| > |a_2| + |c_2|$$

$$|c_3| > |a_3| + |b_3|$$

Diagonally dominant: When the system of equation can be ordered so that each diagonal entry of the coefficient matrix is larger in magnitude than the sum of the magnitude of other coefficients in that row, such a system is called diagonally dominant.

16) Write a algorithm for Gauss-Seidel method.

consider the system of equations

$$a_1x + b_1y + c_1z = d_1 \quad \text{--- (1)}$$

$$a_2x + b_2y + c_2z = d_2 \quad \text{--- (2)}$$

$$a_3x + b_3y + c_3z = d_3 \quad \text{--- (3)}$$

write the above equations in diagonally dominant has

$$|a_1| > |b_1| + |c_1|$$

$$|b_2| > |a_2| + |c_2|$$

$$|c_3| > |a_3| + |b_3|$$

$$\textcircled{1} \Rightarrow x = \frac{1}{a_1} [d_1 - b_1y - c_1z]$$

$$\textcircled{2} \Rightarrow y = \frac{1}{b_2} [d_2 - a_2x - c_2z]$$

$$\textcircled{3} \Rightarrow z = \frac{1}{c_3} [d_3 - a_3x - b_3y]$$

In general,

$$x^{(r+1)} = \frac{1}{a_1} \begin{bmatrix} d_1 - b_1 y^{(r)} - c_1 z^{(r)} \end{bmatrix}$$

$$y^{(r+1)} = \frac{1}{b_2} \begin{bmatrix} d_2 - a_2 x^{(r+1)} - c_2 z^{(r)} \end{bmatrix}$$

$$z^{(r+1)} = \frac{1}{c_3} \begin{bmatrix} d_3 - a_3 x^{(r+1)} - b_3 y^{(r+1)} \end{bmatrix}$$

8m:

17) Solve the following system of equation by Gauss-Jordan method.

$$(i) x_1 + 3x_2 + 3x_3 = 16$$

$$(ii) x_1 + 4x_2 + 3x_3 = 18$$

$$x_1 + 3x_2 + 4x_3 = 19$$

The given equation can be written as $AX=B$

$$\begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 16 \\ 18 \\ 19 \end{bmatrix}$$

The Augmented matrix.

$$[A, B] = \left[\begin{array}{ccc|c} 1 & 3 & 3 & 16 \\ 1 & 4 & 3 & 18 \\ 1 & 3 & 4 & 19 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 3 & 3 & 16 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

$$R_2 \rightarrow R_2 - R_1$$

$$R_3 \rightarrow R_3 - R_1$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 3 & 10 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

$$R_1 \rightarrow R_1 - 3R_3$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array} \right] \quad R_1 \rightarrow R_1 - 3R_3$$

$\therefore$ It is an identity matrix

$$\Rightarrow \left. \begin{array}{l} x = 1 \\ y = 2 \\ z = 3 \end{array} \right\} \parallel \quad \left. \begin{array}{l} x_1 = 1 \\ x_2 = 2 \\ x_3 = 3 \end{array} \right\} \parallel$$

$$\begin{aligned} \text{(ii)} \quad x + y + z &= 9 \\ 2x - 3y + 4z &= 13 \\ 3x + 4y + 5z &= 40 \end{aligned}$$

The given equation can be written as $AX=B$

$$\left[\begin{array}{ccc} 1 & 1 & 1 \\ 2 & -3 & 4 \\ 3 & 4 & 5 \end{array} \right] \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 13 \\ 40 \end{bmatrix}$$

The Augmented matrix

$$[A, B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 2 & -3 & 4 & 13 \\ 3 & 4 & 5 & 40 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 0 & -5 & 2 & -5 \\ 0 & 1 & 2 & 13 \end{array} \right] \quad \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 3R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 0 & -5 & 2 & -5 \\ 0 & 0 & 12 & 60 \end{array} \right] \quad R_3 \rightarrow 5R_3 + R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 0 & -5 & 2 & -5 \\ 0 & 0 & 1 & 5 \end{array} \right] \quad R_3 \rightarrow R_3 \parallel 12$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 0 & 4 \\ 0 & -5 & 0 & -15 \\ 0 & 0 & 1 & 5 \end{array} \right] \begin{array}{l} R_1 \rightarrow R_1 - R_3 \\ R_2 \rightarrow R_2 - 2R_3 \end{array} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 0 & 4 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 5 \end{array} \right] R_2 \rightarrow R_2 / 5$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 5 \end{array} \right] R_1 \rightarrow R_1 - R_2$$

$\therefore$ It is an identity matrix

$$\left. \begin{array}{l} x = 1 \\ y = 3 \\ z = 5 \end{array} \right\} //$$

(iii) $2x_1 + x_2 + x_3 = 10$

$$3x_1 + 2x_2 + 3x_3 = 18$$

$$x_1 + 4x_2 + 9x_3 = 16$$

The given equation can be written as $AX = B$

$$\left[\begin{array}{ccc} 2 & 1 & 1 \\ 3 & 2 & 3 \\ 1 & 4 & 9 \end{array} \right] \left[\begin{array}{c} x_1 \\ x_2 \\ x_3 \end{array} \right] = \left[\begin{array}{c} 10 \\ 18 \\ 16 \end{array} \right]$$

The augmented matrix

$$[A, B] = \left[\begin{array}{ccc|c} 2 & 1 & 1 & 10 \\ 3 & 2 & 3 & 18 \\ 1 & 4 & 9 & 16 \end{array} \right]$$

$$R_1 \leftrightarrow R_3$$

$$\left[\begin{array}{ccc|c} 1 & 4 & 9 & 16 \\ 3 & 2 & 3 & 18 \\ 2 & 1 & 1 & 10 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 4 & 9 & 16 \\ 0 & -10 & -4 & -30 \\ 0 & -7 & -17 & -22 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - 3R_1 \\ R_3 \rightarrow R_3 - 2R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 4 & 9 & 16 \\ 0 & 5 & 12 & 15 \\ 0 & -7 & -17 & -22 \end{array} \right] R_2 \rightarrow R_2 / 5$$

$$\sim \left[\begin{array}{ccc|c} 1 & 4 & 9 & 16 \\ 0 & 5 & 12 & 15 \\ 0 & 0 & -1 & -5 \end{array} \right] R_3 \rightarrow 5R_3 + 7R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 4 & 9 & 16 \\ 0 & 5 & 12 & 15 \\ 0 & 0 & 1 & 5 \end{array} \right] R_3 \rightarrow R_3 / -1$$

$$\sim \left[\begin{array}{ccc|c} 1 & 4 & 9 & 16 \\ 0 & 1 & 12/5 & 3 \\ 0 & 0 & 1 & 5 \end{array} \right] R_2 \rightarrow R_2 / 5$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & -3/5 & 7 \\ 0 & 1 & 12/5 & 3 \\ 0 & 0 & 1 & 5 \end{array} \right] \begin{array}{l} R_1 \rightarrow R_1 + 3/5 R_2 \\ \rightarrow R_2 - 3R_3 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 7 \\ 0 & 1 & 12/5 & 3 \\ 0 & 0 & 1 & 5 \end{array} \right] R_1 \rightarrow R_1 + 3/5 R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 7 \\ 0 & 1 & 0 & -9 \\ 0 & 0 & 1 & 5 \end{array} \right] R_1 \rightarrow R_1 - 12/5 R_3$$

$\therefore$ It is an identity matrix.

$$\left. \begin{array}{l} x = 7 \\ y = -9 \\ z = 5 \end{array} \right\} //$$

$$\left[(5x^2 + 1 - y)P - (988P + 0)Q - 0 \right] \frac{1}{P} = \frac{5x^2 + 1 - y}{P}$$

18) Solve the following system of equations by Gauss-Seidel method correct to 4 decimal places.

$$(i) \quad 9x_1 + 2x_2 + 4x_3 = 10$$

$$2x_1 - 4x_2 + 10x_3 = -15$$

$$x_1 + 10x_2 + 4x_3 = 6$$

Sol: The above equations can be written as diagonally dominant order

$$9x_1 + 2x_2 + 4x_3 = 10 \rightarrow (1)$$

$$x_1 + 10x_2 + 4x_3 = 6 \rightarrow (2)$$

$$2x_1 - 4x_2 + 10x_3 = -15 \rightarrow (3)$$

$$(1) \Rightarrow x_1 = \frac{1}{9} [10 - 2x_2 - 4x_3]$$

$$(2) \Rightarrow x_2 = \frac{1}{10} [6 - x_1 - 4x_3]$$

$$(3) \Rightarrow x_3 = \frac{1}{10} [-15 - 2x_1 + 4x_2]$$

Let $x_2 = x_3 = 0$ (initial value)

I. Iteration:-

$$(i) \quad x_1 = \frac{1}{9} [10] = 1.1111$$

$$(ii) \quad x_2 = \frac{1}{10} [6 - 1.1111 - 4(0)] = 0.4889$$

$$(iii) \quad x_3 = \frac{1}{10} [-15 - 2(1.1111) + 4(0.4889)] = -1.5267$$

II. Iteration:-

$$(i) \quad x_1 = \frac{1}{9} [10 - 2(0.4889) - 4(-1.5267)]$$

$$= 1.6810$$

$$x_2^{(2)} = \frac{1}{10} [6 - 1.6810 - 4(-1.5267)]$$

$$= 1.0426$$

$$x_3^{(2)} = \frac{1}{10} [-15 - 2(1.6810) + 4(1.0426)]$$

$$= -1.4192$$

iii. Iteration:

$$x_1^{(3)} = \frac{1}{9} [10 - 2(1.0426) - 4(-1.4192)]$$

$$= 1.5102$$

$$x_2^{(3)} = \frac{1}{10} [6 - 1.5102 - 4(-1.4192)]$$

$$= 1.0167$$

$$x_3^{(3)} = \frac{1}{10} [-15 - 2(1.5102) + 4(1.0167)]$$

$$= -1.3954$$

iv. Iteration:

$$x_1^{(4)} = \frac{1}{9} [10 - 2(1.0167) - 4(-1.3954)]$$

$$= 1.5054$$

$$x_2^{(4)} = \frac{1}{10} [6 - 1.5054 - 4(-1.3954)]$$

$$= 1.0076$$

$$x_3^{(4)} = \frac{1}{10} [-15 - 2(1.5054) + 4(1.0076)]$$

$$= -1.3980$$

V Iteration:

$$x_1^{(5)} = \frac{1}{9} [10 - 2(1.0076) - 4(-1.3980)]$$

$$= 1.5085$$

$$x_2^{(5)} = \frac{1}{10} [6 - 1.5085 - 4(-1.3980)]$$

$$= 1.0084$$

$$x_3^{(5)} = \frac{1}{10} [-15 - 2(1.5085) + 4(1.0084)]$$

$$= -1.3983$$

VI Iteration:

$$x_1^{(6)} = \frac{1}{9} [10 - 2(1.0084) - 4(-1.3983)]$$

$$= 1.5085$$

$$x_2^{(6)} = \frac{1}{10} [6 - 1.5085 - 4(-1.3983)]$$

$$= 1.0085$$

$$x_3^{(6)} = \frac{1}{10} [-15 - 2(1.5085) + 4(1.0085)]$$

$$= -1.3983$$

VII Iteration:

$$x_1^{(7)} = \frac{1}{9} [10 - 2(1.0085) - 4(-1.3983)]$$

$$= 1.5085$$

$$x_2^{(7)} = \frac{1}{10} [6 - 1.5085 - 4(-1.3983)]$$

$$= 1.0085$$

$$x_3^{(7)} = \frac{1}{10} [-15 - 2(1.5085) + 4(1.0085)]$$

$$= -1.3983$$

$$x_1 = 1.5085$$

$$x_2 = 1.0085$$

$$x_3 = -1.3983$$

Here VI & VII Iteration values are same

$$\therefore x_1 = 1.5085$$

$$x_2 = 1.0085$$

$$x_3 = -1.3983 //$$

$$(iii) \quad x + 5y - z = 10$$

$$x + y + 8z = 20$$

$$4x + 2y + z = 14$$

The above equations can be written as in diagonally dominant order.

$$4x + 2y + z = 14 \rightarrow (1)$$

$$x + 5y - z = 10 \rightarrow (2)$$

$$x + y + 8z = 20 \rightarrow (3)$$

$$(1) \Rightarrow x = \frac{1}{4} [14 - 2y - z]$$

$$(2) \Rightarrow y = \frac{1}{5} [10 - x + z]$$

$$(3) \Rightarrow z = \frac{1}{8} [20 - x - y]$$

Let $y = z = 0$ (Initial values)

Iteration:

$$x = \frac{1}{4} [14] = 3.500$$

$$y = \frac{1}{5} [10 - 3.500] = 1.3$$

$$z = \frac{1}{8} [20 - 3.5 - 1.3] = 1.9$$

II. Iteration :

$$x^{(2)} = \frac{1}{4} [14 - 2(1.3) - 1.9]$$

$$= 2.3750$$

$$y^{(2)} = \frac{1}{5} [10 - 2.3750 + 1.9]$$

$$= 1.9050$$

$$z^{(2)} = \frac{1}{8} [20 - 2.3750 - 1.9050]$$

$$= 1.9650$$

III. Iteration :

$$x^{(3)} = \frac{1}{4} [14 - 2(1.9050) - 1.9650]$$

$$= 2.0563$$

$$y^{(3)} = \frac{1}{5} [10 - 2.0563 + 1.9650]$$

$$= 1.9817$$

$$z^{(3)} = \frac{1}{8} [20 - 2.0563 - 1.9817]$$

$$= 1.9953$$

IV. Iteration

$$x^{(4)} = \frac{1}{4} [14 - 2(1.9817) - 1.9953]$$

$$= 2.0103$$

$$y^{(4)} = \frac{1}{5} [10 - 2.0103 + 1.9953]$$

$$= 1.9970$$

$$z^{(4)} = \frac{1}{8} [20 - 2.0103 - 1.9970]$$

$$= 1.9990$$

$$x^{(5)} = \frac{1}{4} [14 - 2(1.9990) - 1.9991]$$

$$= 2.0017$$

$$y^{(5)} = \frac{1}{5} [10 - 2.0017 + 1.9991]$$

$$= 1.9995$$

$$z^{(5)} = \frac{1}{8} [20 - 2.0017 - 1.9995]$$

$$= 1.9999$$

vi. Iteration

$$x^{(6)} = \frac{1}{4} [14 - 2(1.9995) - 1.9999]$$

$$= 2.0003$$

$$y^{(6)} = \frac{1}{5} [10 - 2.0003 + 1.9999]$$

$$= 1.9999$$

$$z^{(6)} = \frac{1}{8} [20 - 2.0003 - 1.9999]$$

$$= 2$$

vii. Iteration:

$$x^{(7)} = \frac{1}{4} [14 - 2(1.9999) - 2]$$

$$= 2.0001$$

$$y^{(7)} = \frac{1}{5} [10 - 2.0001 + 2]$$

$$= 2$$

$$z^{(7)} = \frac{1}{8} [20 - 2.0001 - 2]$$

$$= 2$$

VII Iteration:

$$x^{(8)} = \frac{1}{4} [14 - 2(2) - 2] = 2$$

$$y^{(8)} = \frac{1}{5} [10 - 2 + 2] = 2$$

$$z^{(8)} = \frac{1}{8} [20 - 2 - 2] = 2$$

VIII Iteration:

$$x^{(9)} = \frac{1}{4} [14 - 2(2) - 2] = 2$$

$$y^{(9)} = \frac{1}{5} [10 - 2 + 2] = 2$$

$$z^{(9)} = \frac{1}{8} [20 - 2 - 2] = 2$$

Here VII & VIII Iteration values are same

$$\therefore \begin{cases} x = 2 \\ y = 2 \\ z = 2 \end{cases}$$

(iii) $x_1 + x_2 + 54x_3 = 1$

$27x_1 + 6x_2 - x_3 = 85$

$6x_1 + 15x_2 + 2x_3 = 72$

The above equation can be written as a diagonally dominant order.

$27x_1 + 6x_2 - x_3 = 85 \rightarrow \textcircled{1}$

$6x_1 + 15x_2 + 2x_3 = 72 \rightarrow \textcircled{2}$

$x_1 + x_2 + 54x_3 = 1 \rightarrow \textcircled{3}$

$$①) x_1 = \frac{1}{27} [85 - 6x_2 + x_3]$$

$$②) x_2 = \frac{1}{15} [72 - 6x_1 + 2x_3]$$

$$③) x_3 = \frac{1}{54} [1 - x_1 - x_2]$$

Let $x_2 = x_3 = 0$ (Initial values)

I. Iteration:

$$x_1 = \frac{1}{27} [85] = 3.1481$$

$$x_2 = \frac{1}{15} [72 - 6(3.1481) - 2(0)] = 3.5408$$

$$x_3 = \frac{1}{54} [1 - 3.1481 - 3.5408] = -0.1054$$

II. Iteration:

$$x_1^{(2)} = \frac{1}{27} [85 - 6(3.5408) - 0.1054] = 2.3574$$

$$x_2^{(2)} = \frac{1}{15} [72 - 6(2.3574) - 2(-0.1054)] = 3.8711$$

$$x_3^{(2)} = \frac{1}{54} [1 - 2.3574 - 3.8711] = -0.0968$$

III. Iteration

$$x_1^{(3)} = \frac{1}{27} [85 - 6(3.8711) + (-0.0968)] = 2.2843$$

$$x_2^{(3)} = \frac{1}{15} [72 - 6(2.2843) - 2(-0.0968)] = 3.8992$$

$$x_3^{(3)} = \frac{1}{54} [1 - 2 \cdot 2843 - 3 \cdot 8992] = -0.0960$$

IV Iteration

$$x_1^{(4)} = \frac{1}{27} [85 - 6(3.8992) - 0.0960] = 2.2781$$

$$x_2^{(4)} = \frac{1}{15} [72 - 6(2.2781) - 2(-0.0960)] = 3.9016$$

$$x_3^{(4)} = \frac{1}{54} [1 - 2 \cdot 2781 - 3 \cdot 9016] = -0.0959$$

V Iteration

$$x_1^{(5)} = \frac{1}{27} [85 - 6(3.9016) - 0.0959] = 2.2776$$

$$x_2^{(5)} = \frac{1}{15} [72 - 6(2.2776) - 2(-0.0959)] = 3.9017$$

$$x_3^{(5)} = \frac{1}{54} [1 - 2 \cdot 2776 - 3 \cdot 9017] = -0.0959$$

VI Iteration

$$x_1^{(6)} = \frac{1}{27} [85 - 6(3.9017) - 0.0959] = 2.2776$$

$$x_2^{(6)} = \frac{1}{15} [72 - 6(2.2776) - 2(-0.0959)] = 3.9017$$

$$x_3^{(6)} = \frac{1}{54} [1 - 2 \cdot 2776 - 3 \cdot 9017] = -0.0959$$

Here ∇ & ∇^* Iteration values are same

$$\begin{aligned} \therefore x_1 &= 2.2776 \\ x_2 &= 3.9017 \\ x_3 &= -0.0959 \end{aligned} \quad \Bigg\} //$$

Gauss-Seidan

10

Identical element in multiplication is 1